MIL-S-17531D

29 July 1970
SUPERSEDING
MIL-S-17531C
21 February 1967

MILITARY SPECIFICATION

SCRAPER, BAKERS

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type and size of table top scraper (see 6.1).

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard
PPP-B-636 - Box, Fiberboard
PPP-B-676 - Boxes, Set-up, Paperboard
PPP-T-45 - Tape, Gummed, Paper

MILITARY

MIL-P-116 - Preservation, Methods Of
MIL-B-121 - Barrier Material, Greaseproofed, Flexible, Waterproofed

STANDARDS

FEDERAL

FED-STD-66 - Steel: Chemical Composition and Hardenability

FSC 7330

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-130 - Identification Marking of U. S. Military Property

MIL-STD-147 - Palletized and Containerized Unit Loads, 40 Inch by 48 Inch Pallets, Skids, Runners, or Pallet Type Base

* DRAWING

ARMY NATICK LABORATORIES

5-11-113 - Scraper, Bakers

(A miniature copy of Drawing 5-11-113, identified herein as figure 1, is attached for information purposes only.)

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, ATTN: Tariff Order Section, 1616 P Street, N. W., Washington, D. C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

Specification E18-67: Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 First article.- This specification contains provisions for first article approval (see 4.2, 6.2, and 6.3).

3.2 Materials.- Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard bakers scraper provided the completed item complies with all provisions of this specification.

3.2.1 Steel sheet and strip.- Steel sheet and strip shall conform to the chemical composition of AISI steel No. 1070 or 1078 of FED-STD-66 and shall be cold finished. The Rockwell hardness shall be not less than 24 on the C scale when tested as specified in 4.3.1.2.

3.2.2 Wood.- Wood used shall be maple, birch, beech, or pecan, conforming to the quality normally used for the purpose in commercial practice.

3.2.3 Rivets.- Rivets shall be brass, conforming to the quality normally used for the purpose in commercial practice and shall be of the compression type.

3.3 Construction.- The scraper construction shall conform to the requirements specified herein and on Drawing 5-11-113. The handle shall be fabricated of wood specified in 3.2.2 and shall be of one-piece construction. The blade shall be constructed of steel specified in 3.2.1 and shall be secured to the handle with either 2 or 3 uniformly spaced rivets specified in 3.2.3. The blade shall fit firmly in the handle slot so that the surfaces of the slot bear evenly against the blade at all points to prevent accumulation of food or foreign matter between the handle and the blade.

3.3.1 Handle and rivets.- After assembly, the rivets and handle shall be machined flush.

3.4 Finish.- The blade shall be polished in accordance with commercial practice. The handle shall be finished in accordance with commercial practice.

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3.5 Marking for identification.- Scrapers shall be marked with the letters "U.S." and the manufacturer's name, trade name, or trademark of such known characters as to be readily identifiable with the manufacturer. Markings shall be applied in accordance with MIL-STD-130 and as shown on Drawing 5-11-113.

3.6 Workmanship.- The finished item shall be clean and well made. All sharp edges, projections, and slivers shall be removed. Handles shall be free of splits, cracks, knots, shakes, worm holes, brashness, and decay.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.- When a preproduction sample is required, it shall be examined for defects listed in table I and dimensions specified. The presence of any defects listed in table I or any dimension not within specified requirements shall be cause for rejection of the preproduction sample.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawing, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.1.1 Steel sheet and strip analysis.- Steel sheet and strip shall be analyzed in accordance with the check analysis procedures of FED-STD-66. Two samples shall be taken from each lot of material unless the lot consists of only one piece, in which case only one sample shall be taken. Failure of any sample to comply shall be cause for rejection of the lot.

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4.3.1.2 Intermediate testing of scraper blade.- The scraper blade shall be tested in accordance with ASTM E18 for conformance with the hardness requirement of 3.2.1. The lot shall be the number of blades offered for inspection at one time. The inspection level shall be S-3 with an acceptable quality level (AQL) of 4.0 defects expressed in terms of defects per hundred units.

4.3.2 End item inspection.- The inspection lot shall be the number of scrapers offered for acceptance at one time. The sample unit shall be one scraper.

4.3.2.1 Visual examination.- Examination of the end item shall be made for defects in table I. The inspection level shall be II with an AQL of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I.- Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Rivets and handle not flush as specified	X	
	Blade not polished; pits, scratches, or tool marks in blade deep enough to hold food particles	X	
	Stain or discoloration on blade		X
Construction and workmanship	Handle wood not one piece construction	X	
	Interior surfaces of handle slot not tight fit against blade to prevent accumulation of food or foreign matter	X	
	Handle not maple, birch, beech, or pecan	X	
	Blade not secured to handle by 2 or 3 compression type rivets	X	
	Rivets not brass		X
	Any sharp edges (except scraping blade) or projection which may cause injury	X	
	Split, crack, knot, shake, worm hole, brashness, or decay in handle	X	
Marking	Omitted, illegible, incomplete		X

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4.3.2.2 Dimensional examination.- Dimensional examination shall be made of the end item to determine compliance with this specification. Any dimension not within the specified requirements shall be classified a defect. The inspection level shall be S-2, with an AQL of 6.5 expressed in terms of defects per hundred units.

4.3.3 Examination of preparation for delivery.- An examination shall be made to determine that preservation, packaging, and marking as required by Section 5 of this specification are complied with. Defects shall be as specified in table II. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of shipping containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

TABLE II.- Examination of preparation for delivery

Examine	Defect
Marking (exterior and interior container)	Omitted, incorrect, illegible; of improper size, location, sequence, or method of application
Materials	Component missing, or damaged
Contents	Number per container is more or less than required
Preservation	Preservative improperly applied or missing

* 4.3.3.1 Examination of shipping containers.- When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing, and strapping shall be in accordance with PPP-B-636.

* 4.3.4 Examination of palletization.- An examination shall be made to determine that palletization is in compliance with Section 5 of this specification. Defects shall be as specified in table III. The sample unit for this examination shall be one palletized unit load ready for shipment. The lot shall be the number of palletized unit loads submitted for inspection at one time. The inspection level shall be S-1 with an AQL of 6.5 expressed in terms of defects per hundred units.

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TABLE III.- Examination of palletization

Examination	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement
Palletization	Not as specified Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.-

5.1.1.1 Preservation.- Each scraper shall be cleaned process C-1, thoroughly dried, and the scraper blade coated with type P-14 preservative in accordance with the requirements of MIL-P-116. The scraper shall then be bagged or completely wrapped with barrier material conforming to type II, grade A, class 2 of MIL-B-121. The bag or wrap shall be secured with gummed paper tape, minimum 1 inch in width conforming to type III, grade A of PPP-T-45, or cotton tape or twine.

5.1.1.2 Packaging.- Twelve preserved, bagged, or wrapped scrapers shall be packaged in the most compact manner in a snug-fitting folding or set-up paperboard box conforming to variety 1, style III, type G, class 1 or style IV, type I or J, class 1 of PPP-B-566, or type I, variety 1, class A, style 2 of PPP-B-676, respectively. Each box shall be closed in accordance with the appendix of the applicable box specification.

5.1.2 Level C.- Scrapers shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice, when it meets these requirements.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

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5.2.1 Level A.- One hundred and forty-four scrapers, preserved and packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V2s of PPP-B-636. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat strapping or tape banding in accordance with the appendix of PPP-B-636.

5.2.2 Level B.- One hundred and forty-four scrapers, preserved and packaged as specified in 5.1, shall be packed in a snug-fitting, fiberboard shipping container conforming to style RSC, type CF, variety optional or type SF, class domestic of PPP-B-636. Closure shall be in accordance with method II of the appendix of PPP-B-636.

5.2.2.1 When specified (see 6.2) the shipping container shall be V3c, V3s, or V4s, fabricated and closed in accordance with PPP-B-636.

5.2.3 Level C.- Scrapers, preserved and packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

* 5.3 Palletization.- Unless otherwise specified (see 6.2) scrapers packed as specified shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The scraper covered by this specification is intended for use in kitchens and galleys of Military installations as a baker's tool for the removal of food particles from work tops and miscellaneous pans.

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article (see 3.1, 4.2, and 6.3).
(c) Selection of applicable levels of preservation and packaging, and packing (see 5.1 and 5.2).
(d) Whether weather-resistant class fiberboard shipping containers are required for level B shipment (see 5.2.2.1).
(e) When palletization is not required (see 5.3).

* 6.3 First article.- The item covered by this specification requires first article approval under the appropriate provisions of paragraph 7-104.55 of the Armed Services Procurement Regulations. The first article should be a preproduction sample and should consist of one baker's scraper. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examination and approval of the first article.

* 6.4 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document, based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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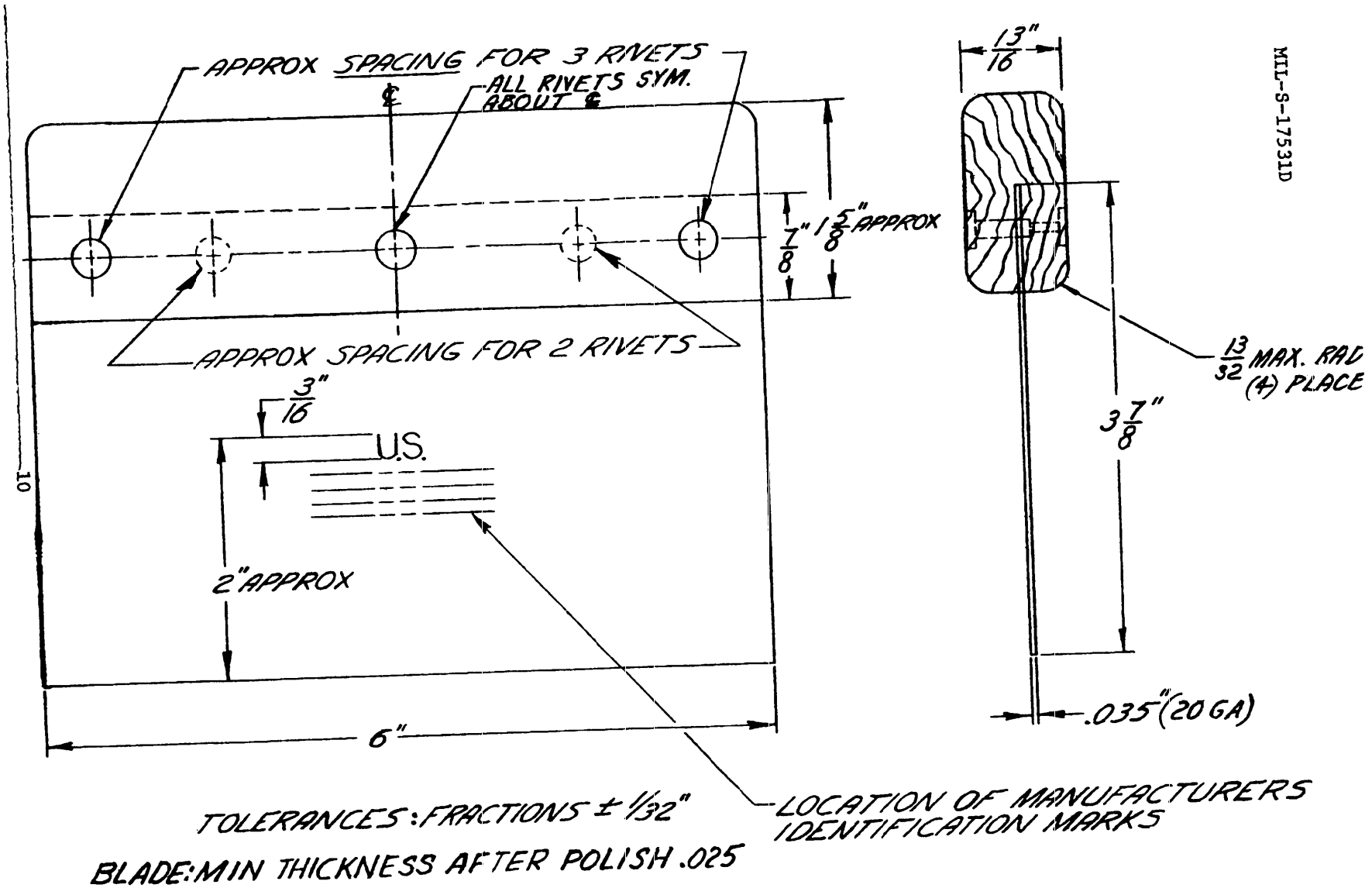


FIGURE-1 SCRAPER, BAKERS'

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION Scraper, Bakers MIL-S-17531D

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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